Percentage Increase and Decrease

Name _____ Date _____ Calculator allowed Version 1 Scan to practise this topic online

KEY WORDS percentage *(ōrau)* · increase *(pikinga)* · decrease *(hekenga)* · multiplier *(tau whakarea)*

A Percentage of an amount ●○○

Remember: Change the percentage to a decimal and multiply: 15% of 80 is $0.15 \times 80 = 12$.

1. Work out.

a) 10% of 70 = _____ **b)** 50% of 46 = _____ **c)** 25% of 80 = _____ **d)** 20% of 35 = _____

e) 15% of 60 = _____ **f)** 5% of 240 = _____ **g)** 30% of $90 = _____ **h)** 12% of $250 = _____

B Multipliers ●○○

Remember: To increase by 20% you need 120% of the amount, so multiply by 1.20. To decrease by 20% you need 80%, so multiply by 0.80.

2. Write the multiplier for each change.

a) Increase by 10% × **b)** Increase by 25% × **c)** Decrease by 10% × **d)** Decrease by 30% ×

e) Increase by 5% × **f)** Decrease by 15% × **g)** Increase by 40% × **h)** Decrease by 8% ×

C Increasing an amount ●●○

3. Increase each amount by the percentage given.

a) 50 by 10% **b)** 80 by 25% **c)** 120 by 5%

d) 200 by 15% **e)** 64 by 50% **f)** 350 by 20%

g) $90 by 30% **h)** $45 by 12% **i)** $1,500 by 4%

D Decreasing an amount ●●○

4. Decrease each amount by the percentage given.

a) 50 by 10% **b)** 80 by 25% **c)** 120 by 5%

d) 200 by 15% **e)** 64 by 50% **f)** 350 by 20%

g) $90 by 30% **h)** $45 by 12% **i)** $1,500 by 4%

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E Discounts and mark-ups ●●○

Remember: A discount is a percentage decrease in the price. A mark-up is a percentage increase.

5. Find the new price.

a) A jacket costs $80. It is on sale at 25% off.

b) A bike costs $240. There is a discount of 15%.

c) A shop buys a toy for $36 and adds a mark-up of 40%.

d) A cafe pays $65 for a box of fruit and adds a mark-up of 20%.

F Finding the original amount ●●●

Watch out: To go back to the original amount, divide by the multiplier. Do not take the percentage off the new amount.

6. Find the original amount.

a) After a 10% increase, an amount is 55.

b) After a 20% decrease, an amount is 64.

c) A price rises by 25% to $90.

d) In a 30% off sale, a shirt costs $42.

G Problems ●●●

7. Rent of $1,200 a month goes up by 5%. Find the new rent.

8. A car bought for $18,000 loses 15% of its value in a year. What is it worth after one year?

9. Hemi earns $25 an hour. He gets a pay rise of 6%. Find his new hourly pay.

★ Extension

10. Find the percentage change. Say whether it is an increase or a decrease.

a) from 40 to 50 **b)** from 80 to 60 **c)** from $250 to $290

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Percentage Increase and Decrease

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

KEY WORDS percentage *(ōrau)* · increase *(pikinga)* · decrease *(hekenga)* · multiplier *(tau whakarea)*

A Percentage of an amount ●○○

Remember: Change the percentage to a decimal and multiply: 15% of 80 is $0.15 \times 80 = 12$.

1. Work out.

a) 10% of 90 = _____ **b)** 50% of 58 = _____ **c)** 25% of 60 = _____ **d)** 20% of 45 = _____

e) 15% of 40 = _____ **f)** 5% of 320 = _____ **g)** 30% of \$70 = _____ **h)** 12% of \$150 = _____

B Multipliers ●○○

Remember: To increase by 20% you need 120% of the amount, so multiply by 1.20. To decrease by 20% you need 80%, so multiply by 0.80.

2. Write the multiplier for each change.

a) Increase by 20% × **b)** Increase by 15% × **c)** Decrease by 20% × **d)** Decrease by 40% ×

e) Increase by 8% × **f)** Decrease by 25% × **g)** Increase by 60% × **h)** Decrease by 5% ×

C Increasing an amount ●●○

3. Increase each amount by the percentage given.

a) 70 by 10% **b)** 60 by 25% **c)** 140 by 5%

d) 300 by 15% **e)** 86 by 50% **f)** 450 by 20%

g) \$70 by 30% **h)** \$55 by 12% **i)** \$2,500 by 4%

D Decreasing an amount ●●○

4. Decrease each amount by the percentage given.

a) 70 by 10% **b)** 60 by 25% **c)** 140 by 5%

d) 300 by 15% **e)** 86 by 50% **f)** 450 by 20%

g) \$70 by 30% **h)** \$55 by 12% **i)** \$2,500 by 4%

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E Discounts and mark-ups ●●○

Remember: A discount is a percentage decrease in the price. A mark-up is a percentage increase.

5. Find the new price.

a) A jacket costs $90. It is on sale at 20% off.

b) A bike costs $360. There is a discount of 15%.

c) A shop buys a toy for $48 and adds a mark-up of 25%.

d) A cafe pays $75 for a box of fruit and adds a mark-up of 40%.

F Finding the original amount ●●●

Watch out: To go back to the original amount, divide by the multiplier. Do not take the percentage off the new amount.

6. Find the original amount.

a) After a 10% increase, an amount is 77.

b) After a 20% decrease, an amount is 96.

c) A price rises by 25% to $60.

d) In a 30% off sale, a shirt costs $56.

G Problems ●●●

7. Rent of $1,500 a month goes up by 4%. Find the new rent.

8. A car bought for $24,000 loses 12% of its value in a year. What is it worth after one year?

9. Hemi earns $28 an hour. He gets a pay rise of 5%. Find his new hourly pay.

★ Extension

10. Find the percentage change. Say whether it is an increase or a decrease.

a) from 50 to 60 **b)** from 60 to 45 **c)** from $200 to $236

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